

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250283612

Kind Code

A1

Publication Date

September 11, 2025

Inventor(s)

PARK; Sangjin et al.

DEHUMIDIFIER

Abstract

A dehumidifier is provided that may include a base with an external air intake on one side thereof; a case disposed on an upper surface of the base and having a suction grille formed on a rear surface thereof; a top cover disposed on an upper surface of the case and having a discharge port; a discharge cover that opens and closes the discharge port; a drain pan including a first portion that extends laterally to divide an internal space of the case into an upper space and a lower space, and a second portion that extends from a lower surface of the first portion and disposed on the base; a water tank disposed on the upper surface of the base and that stores condensed water generated during a dehumidification process; a compressor disposed on the upper surface of the base spaced apart from the water tank; a fan module disposed on the first portion and having an outlet that communicates with the discharge port of the top cover; a condenser mounted on the first portion at an intake side of the fan module; an evaporator mounted on the first portion behind the condenser; and a control box coupled to the second portion and disposed directly above the external air intake. The fan module may include a fan housing disposed on the first portion, and a fan accommodated in the fan housing. A dripping hole may be formed in the fan housing.

Inventors: PARK; Sangjin (Seoul, KR), YOON; Juhyup (Seoul, KR), BEEN; Jin Woo (Seoul, KR), LEE; Hyunwoo (Seoul, KR)

Applicant: LG ELECTRONICS INC. (Seoul, KR)

Family ID: 1000008008635

Appl. No.: 18/772551

Filed: July 15, 2024

Foreign Application Priority Data

KR 10-2024-0033872

Mar. 11, 2024

Publication Classification

Int. Cl.: F24F1/0083 (20190101); F24F13/20 (20060101); F24F13/22 (20060101)

U.S. Cl.:

CPC F24F1/0083 (20190201); F24F13/20 (20130101); F24F13/22 (20130101);

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S)

[0001] This application claims the benefits of priority to Korean Patent Application No. 10-2024-0033872, filed in Korea on Mar. 11, 2024, the contents of which is incorporated herein by reference in its entirety.

BACKGROUND

1. Field

[0002] A dehumidifier is disclosed herein.

2. Background

[0003] A dehumidifier is a device that lowers a humidity in a space in which the dehumidifier is installed by suctioning in air, removing moisture contained in the air, and discharging the dehumidified air back into the room. In general, dehumidifiers are widely used to lower the humidity of indoor air in humid summer or to quickly dry laundry.

[0004] Dehumidifiers are divided into a desiccant type and a condensation type depending on a dehumidification method. In the condensation type, a refrigerant cycle is installed inside of the dehumidifier, air suctioned in by a fan passes through an evaporator to condense moisture contained in the air, and then passes through the condenser and is discharged back into a room in a heated state.

[0005] In the case of an inverter dehumidifier, electrical components that generate a lot of heat are installed inside of a control box, and it is necessary to quickly cool the electrical components that generate a lot of heat. However, conventional dehumidifiers do not provide a separate cooling passage structure to cool the control box, so there is a high risk of overheating the control box.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] Embodiments will be described in detail with reference to the following drawings in which like reference numerals refer to like elements, and wherein:

[0007] FIG. 1 is a front perspective view of a dehumidifier according to an embodiment;

[0008] FIG. 2 is a rear perspective view of the dehumidifier of FIG. 1;

[0009] FIG. 3 is a schematic diagram showing an internal configuration of the dehumidifier according to an embodiment;

[0010] FIG. 4 is a perspective view of a base of a housing of the dehumidifier according to an embodiment;

[0011] FIG. 5 is a longitudinal cross-sectional view of the base, taken along line V-V in FIG. 4; and

[0012] FIGS. 6 to 8 are schematic diagrams showing a flow of indoor air flowing in through an external air inlet of the base.

DETAILED DESCRIPTION

[0013] Hereinafter, a dehumidifier according to an embodiment will be described with reference to the drawings.

[0014] FIG. 1 is a front perspective view of a dehumidifier according to an embodiment. FIG. 2 is a

rear perspective view of the dehumidifier of FIG. 1. FIG. 3 is a schematic diagram showing an internal configuration of the dehumidifier according to an embodiment.

[0015] Referring to FIGS. 1 to 3, dehumidifier **10** according to an embodiment may have a slim shape with a relatively thin thickness in a frontward-to-backward direction. The dehumidifier **10** may have a thickness extending in the frontward-to-backward direction, a width extending in a leftward-rightward or lateral direction, and a height extending in a vertical direction.

[0016] The dehumidifier **10** may include a housing that forms an exterior thereof. The dehumidifier **10** may further include a dehumidification module installed inside of the housing. The dehumidifier **10** may further include a water tank **20** that stores condensate generated during the dehumidification process.

[0017] The housing may include a base **11**, a case **12**, and a top cover **13**. The base **11** may form a lower portion of the dehumidifier **10**. A plurality of wheels **16** may be mounted on a bottom of the base **11** to facilitate movement of the dehumidifier **10**. The case **12** may be disposed on an upper edge of the base **11**, and an upper opening of the case **12** may be shielded by the top cover **13**.

[0018] A suction grill **121** may be formed on a back of the case **12** to intake indoor air. A filter module may be mounted in front of the suction grill **121**. The suction grill **121** may be defined as a suction port or inlet through which indoor air is suctioned.

[0019] A main discharge port **131**, through which dehumidified air may be discharged may be formed on one or a first side of the top cover **13**. The main discharge port **131** may be selectively opened or closed by the discharge cover **14**.

[0020] A handle **15** may be provided on the other or a second side of the top cover **13**. The handle **15** has an n-shape so that a vertical portion thereof may be inserted into the case **12** or pulled out from the case **12**. When the vertical portion is fully inserted into the case **12**, a horizontal portion of the handle **15**, that is, an upper surface of a portion configured to be held by a hand, may be flush with the top cover **13**.

[0021] The main discharge port **131** may be disposed at one or a first edge of the top cover **13**. The handle **15** may be placed on the other or a second edge of the top cover **13**. The main discharge port **131** and the handle **15** may be located on opposite edges. A discharge grill **141** may be mounted on the main discharge port **131**.

[0022] A cutout into which the water tank **20** may be inserted may be formed at one edge of the case **12**, and the water tank **20** may be detachable from the case **12**. When the water tank **20** is mounted on the case **12**, an outer surface of the water tank **20** may form a portion of the case **12**. The water tank **20** may be located right beneath the main discharge port **131**.

[0023] The dehumidification module may include a compressor **31** that compresses a refrigerant, a condenser **32** that condenses the refrigerant that has passed through the compressor **31**, an expansion valve that expands the refrigerant that has passed through the condenser **32** into a two-phase refrigerant, and an evaporator **33** that evaporates the refrigerant that has passed through the expansion valve. The dehumidification module may further include a fan module **34** that suctions indoor air through the suction grill **121**. The evaporator **33** may be located in front of the suction grill **121**, the condenser **32** may be located in front of the evaporator **33**, and the fan module **34** may be located in front of the condenser **32**.

[0024] The condenser **32** and the evaporator **33** may be seated on an upper surface of a drain pan **35**. Condensed water flowing down from a surface of the evaporator **33** may flow along the drain pan **35** and collected in the water tank **20**.

[0025] An outlet of the fan module **34** may be connected to the main discharge port **131**.

[0026] The air introduced into the case **12** through the suction grill **121** may be dehumidified while passing through the evaporator **33**, and moisture contained in the air condensed. As a result, the incoming air becomes low-temperature and dry as it passes through the evaporator **33**.

[0027] The low-temperature dry air absorbs heat while passing through the condenser **32**, and as a result, is changed to a high-temperature dry state and is then discharged back into the room through

the main discharge port **131**. The term “high temperature air” may refer to air with a temperature slightly higher than a temperature of an indoor space in which the dehumidifier **10** is installed.

[0028] A control box **17** may be accommodated inside of the case **12**, to control operation of the dehumidification module and various drive means.

[0029] A sub discharge port **122** through which dehumidified air may be discharged may be further formed in the case **12**. The sub discharge port **122** may be selectively opened or closed by a damper module **40**.

[0030] When the sub discharge port **122** is open, the main discharge port **131** may be closed, and when the sub discharge port **122** is closed, the main discharge port **131** may be open. Accordingly, the dehumidified air may be selectively discharged only through either the main discharge port **131** or the sub discharge port **122**.

[0031] The sub discharge port **122** may be formed on a front surface or a rear surface of the case **12**. The sub discharge port **122** may be disposed at an upper side of the fan module **34**. For example, the sub discharge port **122** may be formed on the rear surface of the case **12** and may be located above the suction grill **121**.

[0032] The dehumidifier **10** may further include the damper module **40** that selectively opens or closes the sub discharge port **122**. The damper module **40** may be installed inside of the sub discharge port **122** and may open or close the sub discharge port **122**. The damper module **40** may operate in a direction to close the sub discharge port **122**.

[0033] FIG. **4** is a perspective view of a base of a housing of the dehumidifier according to an embodiment. FIG. **5** is a longitudinal cross-sectional view of the base, taken along V-V in FIG. **4**.

[0034] Referring to FIGS. **4** and **5**, an upper surface of the base **11** of the dehumidifier **10** according to an embodiment may include a water tank seating portion **111**, a compressor seating portion **112**, and a control box seating portion **113**. An external air inlet **114** may be formed on or at one side of the base **11** corresponding to the control box seating portion **113**. External air may refer to air outside of the dehumidifier **10**, that is, indoor air of an indoor space in which the humidifier **10** is disposed.

[0035] The external air inlet **114** may be provided in the shape of a hole that penetrates the base **11** or an external air inlet sleeve with an open upper surface as shown in the drawings. The external air inlet **114** may be located right below the control box **17**, and may be formed at a position spaced apart from a bottom surface of the control box **17**.

[0036] FIGS. **6** to **8** are schematic diagrams showing a flow of indoor air flowing in through an external air inlet of the base. Referring to FIG. **6**, some of the indoor air flowing into the dehumidifier **10** through the external air inlet **114** flows into the control box **17** through an opening formed on the bottom surface of the control box **17**, and a remainder flows upwards along bottom and side surfaces of the control box **17**. That is, the air cools the control box **17** while flowing up along a narrow space formed between the case **12** and the control box **17**. Additionally, a circuit board **173** on which a number of electrical components are mounted may be installed inside of the control box **17**, and in particular, a heat sink **171** may be mounted on the heating component **172**.

[0037] Additionally, the heat sink **171** may penetrate the control box **17** and protrude out of the control box **17**. That is, the heat sink **171** may be exposed to the narrow space between the case **12** and the control box **17** and may directly exchange heat with air that flows upwards after flowing into the external air inlet **114**. Indoor air flowing in through the external air inlet **114** may cool the heat sink **171** as it flows upwards along the space formed between the control box **17** and the case **12**.

[0038] Referring to FIG. **7**, the indoor air that flows upwards while cooling the control box **17** may flow towards the drain pan **35** and flow into a gap between the fan module **34** and the drain pan **35**. The drain pan **35** may include a first portion **351** that laterally extends so that a fan housing of the fan module **34** may be seated thereon, and second portions **352** that extend downward from a bottom of the first portion **351** to be located on the base **11**.

[0039] The second portions **352** may extend from one or a first edge of the first portion **351** and a point spaced apart from the first edge towards the other or a second side, respectively. The control box **17** may be coupled to a second portion that extends from the first edge of the first portion **351** among the second portions **352**, and the second portion that extends from the point spaced from the first edge towards the second side is configured to surround a side surface of the water tank **20**.

[0040] In addition, an air guide portion (air guide) **353** may be formed at one edge of the first portion **351**, in particular, at the edge of the first portion **351** corresponding to an upper end of the second portion **352** on which the control box **17** is mounted. Air flowing up along the surface of the control box **17** flows up towards the fan module **34** through the air guide portion **353**.

[0041] Referring to FIG. **8**, the fan module **34** may include the fan housing and a fan accommodated inside of the fan housing. A dripping hole **341** may be formed at a lower end of the fan housing. During the dehumidification process, condensed water condensed on a surface of the fan and an inner surface of the fan housing may be discharged to the drain pan **35** through the dripping hole **341**.

[0042] The air that flows upward while cooling the control box **17** flows into the fan housing through the dripping hole **341**. The air introduced into the fan housing may be discharged back into the room through the main discharge port **131** while evaporating some of the condensate condensed on the fan and the fan housing, when the fan rotates.

[0043] By forming the dripping hole **341**, when the fan rotates, negative pressure is created outside of the fan housing, allowing indoor air to flow into the dehumidifier **10** through the external air inlet **114**. In addition, the indoor air flowing into the dehumidifier **10** through the external air inlet **114** cools the control box **17** as it flows upward, and absorbs heat while passing through the control box **17**, increasing its temperature.

[0044] In addition, the indoor air with an increased temperature flows into the fan module **34** through the dripping hole **341**, and also performs the function of evaporating condensation water formed inside of the fan module **34**. The air flowing into the fan module **34** through the dripping hole **341** dries the inside of the fan module **34**, thereby preventing fungus from growing inside of the fan module **34**.

[0045] Embodiments disclosed herein provide a dehumidifier that may include a base with an external air intake on one side thereof; a case erected on an edge of an upper surface of the base and having a suction grille formed on a rear surface thereof; a top cover placed on an upper surface of the case and having a discharge port; a discharge cover that opens or closes the discharge port; a drain pan including a first part or portion that extends laterally to divide an internal space of the case into an upper space and a lower space, and a second part or portion that extends from a lower surface of the first part and placed on the base; a water tank placed on one side of the upper surface of the base and that stores condensed water generated during a dehumidification process; a compressor placed on one side of the upper surface of the base spaced apart from the water tank; a fan module placed on the first portion and having an outlet that communicates with the discharge port of the top cover; a condenser mounted on the first portion to be located on an intake side of the fan module; an evaporator mounted on the first portion to be located behind the condenser; and a control box coupled to the second part and erected at a point right above the external air intake. The fan module may include a fan housing placed on the first portion; and a fan accommodated in the fan housing. A dripping hole may be formed in the fan housing.

[0046] The external air inlet may be formed on an upper surface of a sleeve that extends upward from the base. A separated space may be formed between the case and the control box so that indoor air flowing into the case through the external air inlet may flow upward while contacting the control box. An air guide portion may be formed between an edge of the first portion and the case, such that the air flowing upward along the surface of the control box may be guided to an upper side of the first portion. The dripping hole may be formed at a lower edge of the fan housing.

[0047] According to embodiments disclosed herein, a hole through which indoor air flows may be

formed at a bottom of the dehumidifier where the control box is located, such that indoor air flowing in through the hole may flow upward to cool the control box, and thereafter flow into the fan module through the dripping hole formed in the fan module. As a result, electrical components that generate a lot of heat may be cooled quickly, thereby preventing overheating of the control box.

[0048] It will be understood that when an element or layer is referred to as being “on” another element or layer, the element or layer can be directly on another element or layer or intervening elements or layers. In contrast, when an element is referred to as being “directly on” another element or layer, there are no intervening elements or layers present. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items.

[0049] It will be understood that, although the terms first, second, third, etc., may be used herein to describe various elements, components, regions, layers and/or sections, these elements, components, regions, layers and/or sections should not be limited by these terms. These terms are only used to distinguish one element, component, region, layer or section from another region, layer or section. Thus, a first element, component, region, layer or section could be termed a second element, component, region, layer or section without departing from the teachings of the present invention.

[0050] Spatially relative terms, such as “lower”, “upper” and the like, may be used herein for ease of description to describe the relationship of one element or feature to another element(s) or feature(s) as illustrated in the figures. It will be understood that the spatially relative terms are intended to encompass different orientations of the device in use or operation, in addition to the orientation depicted in the figures. For example, if the device in the figures is turned over, elements described as “lower” relative to other elements or features would then be oriented “upper” relative to the other elements or features. Thus, the exemplary term “lower” can encompass both an orientation of above and below. The device may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein interpreted accordingly.

[0051] The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the invention. As used herein, the singular forms “a”, “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “comprises” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

[0052] Embodiments are described herein with reference to cross-section illustrations that are schematic illustrations of idealized embodiments (and intermediate structures). As such, variations from the shapes of the illustrations as a result, for example, of manufacturing techniques and/or tolerances, are to be expected. Thus, embodiments should not be construed as limited to the particular shapes of regions illustrated herein but are to include deviations in shapes that result, for example, from manufacturing.

[0053] Unless otherwise defined, all terms (including technical and scientific terms) used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this invention belongs. It will be further understood that terms, such as those defined in commonly used dictionaries, should be interpreted as having a meaning that is consistent with their meaning in the context of the relevant art and will not be interpreted in an idealized or overly formal sense unless expressly so defined herein.

[0054] Any reference in this specification to “one embodiment,” “an embodiment,” “example embodiment,” etc., means that a particular feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment of the invention. The appearances of such phrases in various places in the specification are not necessarily all referring to the same embodiment. Further, when a particular feature, structure, or characteristic is described in

connection with any embodiment, it is submitted that it is within the purview of one skilled in the art to effect such feature, structure, or characteristic in connection with other ones of the embodiments.

[0055] Although embodiments have been described with reference to a number of illustrative embodiments thereof, it should be understood that numerous other modifications and embodiments can be devised by those skilled in the art that will fall within the spirit and scope of the principles of this disclosure. More particularly, various variations and modifications are possible in the component parts and/or arrangements of the subject combination arrangement within the scope of the disclosure, the drawings and the appended claims. In addition to variations and modifications in the component parts and/or arrangements, alternative uses will also be apparent to those skilled in the art.

Claims

1. A dehumidifier, comprising: a base with an external air intake on one side thereof; a case disposed on an upper surface of the base and having a suction grille formed on a rear surface thereof; a top cover disposed on an upper surface of the case and having a discharge port; a discharge cover that opens and closes the discharge port; a drain pan including a first portion that extends laterally to divide an internal space of the case into an upper space and a lower space, and a second portion that extends from a lower surface of the first portion and disposed on the base; a water tank disposed on the upper surface of the base and that stores condensed water generated during a dehumidification process; a compressor disposed on the upper surface of the base spaced apart from the water tank; a fan module disposed on the first portion and having an outlet that communicates with the discharge port of the top cover; a condenser mounted on the first portion at an intake side of the fan module; an evaporator mounted on the first portion behind the condenser; and a control box coupled to the second portion and disposed directly above the external air intake, wherein the fan module includes: a fan housing disposed on the first portion; a fan accommodated in the fan housing, and wherein a dripping hole is formed in the fan housing.
2. The humidifier according to claim 1, wherein the external air intake is formed at an upper surface of a sleeve that extends upward from the base.
3. The humidifier according to claim 2, wherein a space is formed between the case and the control box so that indoor air flowing into the case through the external air intake flows upward along the control box.
4. The humidifier according to claim 3, wherein an air guide is formed between an edge of the first portion and the case, such that the air flowing upward along the surface of the control box is guided to an upper side of the first portion.
5. The humidifier according to claim 4, wherein the dripping hole is formed at a lower edge of the fan housing.
6. A dehumidifier, comprising: a base with an external air intake; a case disposed on the base and having a suction inlet formed on a surface thereof; a top cover disposed on the case and having a discharge outlet; a discharge cover that opens and closes the discharge outlet; a drain pan including a first portion that extends laterally to divide an internal space of the case into an upper space and a lower space, and a second portion that extends downward from the first portion to the base; a water tank disposed on the base and that stores condensed water generated during a dehumidification process; a compressor disposed on the base spaced apart from the water tank; a fan module disposed on the first portion and having an outlet that communicates with the discharge outlet of the top cover; a condenser mounted on the first portion at an intake side of the fan module; an evaporator mounted on the first portion behind the condenser; and a control box coupled to the second portion and disposed directly above the external air intake, wherein the fan module includes: a fan housing disposed on the first portion; a fan accommodated in the fan housing,

- wherein a space is formed between the case and the control box so that indoor air flowing into the case through the external air intake flows upward along the control box.
- 7.** The humidifier according to claim 6, wherein the external air intake is formed at an upper surface of a sleeve that extends upward from the base.
- 8.** The humidifier according to claim 6, wherein an air guide is formed between an edge of the first portion and the case, such that the air flowing upward along the surface of the control box is guided to an upper side of the first portion.
- 9.** The humidifier according to claim 6, wherein the dripping hole is formed at a lower edge of the fan housing.
- 10.** A dehumidifier, comprising: a base with an external air intake that extends upward therefrom; a case disposed on the base and having a suction inlet formed on a surface thereof; a top cover disposed on an upper surface of the case and having a discharge outlet; a discharge cover that opens and closes the discharge outlet; a drain pan including a first portion that extends laterally to divide an internal space of the case into an upper space and a lower space, and a second portion that extends downward from a lower surface of the first portion to the base; a water tank disposed on the base and that stores condensed water generated during a dehumidification process; a compressor disposed on the base spaced apart from the water tank; a fan module disposed on the first portion and having an outlet that communicates with the discharge outlet of the top cover; a condenser mounted on the first portion at an intake side of the fan module; an evaporator mounted on the first portion behind the condenser; and a control box coupled to the second portion and disposed directly above the external air intake, wherein the fan module includes: a fan housing disposed on the first portion; a fan accommodated in the fan housing, wherein a dripping hole is formed in the fan housing through which condensed water condensed on a surface of the fan and an inner surface of the fan housing is discharged to the drain pan, and wherein a space is formed between the case and the control box so that indoor air flowing into the case through the external air intake flows upward along the control box.
- 11.** The humidifier according to claim 10, wherein the external air intake includes a sleeve that extends upward from the base.
- 12.** The humidifier according to claim 10, wherein an air guide is formed between an edge of the first portion and the case, such that the air flowing upward along the surface of the control box is guided to an upper side of the first portion.
- 13.** The humidifier according to claim 10, wherein the dripping hole is formed at a lower edge of the fan housing.
-